

NUT 201: Principles of Nutrition and Dietetics

I. Introduction to Nutrition

Nutrition is the science that studies how the body utilizes food for growth, development, and maintenance.

Essential Nutrients: Nutrients are substances in food that the body needs for growth, repair, and energy. They are classified into two categories:

II. Macronutrients: (Needed in large amounts)

Carbohydrates

Sources

Grains (bread, rice, pasta)

Fruits and vegetables

Legumes

Dairy products

Simple carbohydrates: Sugars, found in fruits, milk, and processed foods.

Complex carbohydrates: Starches and fibre, found in whole grains, vegetables, and legumes. Glycaemic index (GI): A measure of how quickly a carbohydrate is digested and absorbed, affecting blood sugar levels.

Functions

- ◇ Primary energy source
- ◇ Provide 4 calories per gram
- ◇ Support brain and nervous system function
- ◇ Aid in protein sparing

Digestion and Absorption

- ◇ Begins in the mouth with salivary amylase (enzyme)

- ◇ Continued in small intestine by pancreatic amylase
- ◇ Broken down into simple sugars (glucose, fructose, galactose)
- ◇ Absorbed through small intestine walls into bloodstream

Potential Problems

- ◇ Excessive intake can lead to obesity
- ◇ Insulin resistance: condition where cells become less responsive to insulin, cells therefore can't absorb glucose
- ◇ Diabetes: A metabolic disorder characterized by high blood sugar levels
- ◇ Metabolic syndrome

2. Proteins: Composed of amino acids, essential for building and repairing tissues.

Complete proteins contain all essential amino acids, while incomplete proteins lack one or more.

Sources

- ◇ Meat, poultry, fish
- ◇ Eggs
- ◇ Dairy products
- ◇ Legumes
- ◇ Nuts and seeds

Functions

- ◇ Build and repair body tissues
- ◇ Enzyme and hormone production
- ◇ Immune system support
- ◇ Provide 4 calories per gram
- ◇ Transport molecules in the body

Digestion and Absorption

- ◇ Protein digestion begins in the stomach with pepsin
- ◇ Continued in small intestine by pancreatic enzymes
- ◇ Broken down into amino acids
- ◇ Absorbed through small intestine walls

Potential Problems

- ◇ Protein malnutrition
- ◇ Amino acid deficiencies
- ◇ Kidney strain from excessive protein intake

3. Lipids (Fats): Saturated fats: Found in animal products and some plant oils, linked to heart disease.

Unsaturated fats: Found in plant-based oils, considered healthier.

Trans fats: Artificially created fats, linked to heart disease and other health problems.

Sources

- Oils
- Butter
- Fatty fish
- Nuts and seeds
- Avocados
- Animal products

Functions

- Energy storage
- Hormone production
- Cell membrane structure
- Provide 9 calories per gram
- Insulation and protection of organs
- Absorption of fat-soluble vitamins

Digestion and Absorption

- Emulsified by bile in small intestine
- Broken down by lipase enzymes
- Converted to fatty acids and glycerol
- Absorbed as chylomicrons through lymphatic system

(lymphatic system - network of tissues, vessels, and organs that helps body remove toxins, waste, and other unwanted materials)

Potential Problems

- Cardiovascular diseases
- Obesity
- Atherosclerosis
- Cholesterol imbalances

III. Micronutrients

1. Vitamins

Major Vitamin Groups: Fat-soluble vitamins: Vitamins A, D, E, and K, stored in the body's fat tissues.

Water-soluble vitamins: Vitamins B and C, not stored in the body and need to be replenished regularly.

Key Functions

- Metabolic regulation
- Antioxidant protection
- Growth and development
- Immune system support

Potential Deficiency Issues

- ◇ Night blindness (Vitamin A)
- ◇ Rickets (Vitamin D)
- ◇ Bleeding disorders (Vitamin K)
- ◇ Scurvy (Vitamin C)

2. Minerals: Major minerals: Needed in larger amounts, such as calcium, phosphorus, magnesium, sodium, potassium, and chloride.

Trace minerals: Needed in smaller amounts, such as iron, zinc, copper, iodine, selenium, and fluoride.

Essential Minerals

- ◇ Calcium
- ◇ Iron
- ◇ Zinc
- ◇ Magnesium
- ◇ Potassium
- ◇ Sodium

Functions

- ◇ Bone and teeth formation
- ◇ Nerve transmission
- ◇ Muscle contraction
- ◇ Enzyme activation
- ◇ Fluid balance

Potential Problems

- Anemia (Iron deficiency)
- Osteoporosis (Calcium deficiency)
- Electrolyte imbalances

Dietary Guidelines

Balance: Consume a variety of foods from all food groups.

Moderation: Limit intake of unhealthy foods and drinks.

Variety: Choose different foods within each food group to ensure adequate nutrient intake.

Energy balance: Consume the amount of energy needed to maintain a healthy weight.

Hydration: Drink plenty of water throughout the day.

IV. Energy Balance Principles

1. Basal Metabolic Rate (BMR)

Energy required for basic life functions

Influenced by:

- Age
- Gender
- Body composition
- Genetics

2. Caloric Intake and Expenditure

Energy balance equation:

Calories In - Calories Out = Weight Change

Weight maintenance requires balanced intake and expenditure

3. Factors Affecting Energy Expenditure

Physical activity

Thermic effect of food

Non-exercise activity thermogenesis (NEAT)

V. Nutritional Considerations

1. Dietary Guidelines

Balanced diet

Variety of food sources

Portion control

Adequate hydration

5. Nutrition and Health

Chronic diseases: Poor nutrition is a major risk factor for chronic diseases such as heart disease, stroke, diabetes, and cancer.

Immune function: Adequate nutrition is essential for a strong immune system.

Mental health: Nutrient deficiencies can affect mood, concentration, and memory.

Weight management: Balanced nutrition and regular physical activity are key to maintaining a healthy weight.

6. Special Populations

Infants and children: Need nutrient-dense foods for growth and development.

Pregnant and lactating women: Have increased nutrient needs to support foetal and infant growth.

Older adults: May have decreased nutrient needs but require nutrient-dense foods to prevent deficiencies.

Athletes: Have increased energy and nutrient needs to support physical activity.

Conclusion

Nutrition plays a crucial role in maintaining overall health and well-being. By understanding the basic principles of nutrition and making informed food choices, individuals can improve their health and reduce their risk of chronic diseases

NUTRIENT DEFICIENCY DISEASES

Protein Deficiency Diseases

Kwashiorkor: Characterized by oedema, muscle wasting, stunted growth

Marasmus: Severe calorie and protein deficiency, leading to extreme weight loss and muscle breakdown

Vitamin Deficiency Diseases:

Vitamin A: Night blindness, impaired immune function

Vitamin B1 (Thiamine): Beriberi, neurological complications

Vitamin B3 (Niacin): Pellagra, causing dermatitis, diarrhoea, dementia

Vitamin B12: Pernicious anaemia, neurological damage

Vitamin C: Scurvy, poor wound healing, weakened connective tissue

Vitamin D: Rickets in children, osteomalacia in adults

Vitamin E: Rare, but can cause neurological problems

Vitamin K: Impaired blood clotting

Mineral Deficiency Diseases:

Iron: Anaemia, reduced oxygen transport

Iodine: Goitre, impaired thyroid function

Calcium: Osteoporosis, weak bones

Zinc: Growth retardation, impaired immune function

Magnesium: Muscle weakness, cardiac issues

Primary risk factors include poverty, limited diet diversity, malnutrition, and chronic health conditions.

Digestion, Absorption, and Nutrient Functions

Digestion Process

1. Mechanical Breakdown

Begins in mouth through chewing

Continues in stomach via muscular contractions

Breaks food into smaller particles

Increases surface area for chemical digestion

2. Chemical Digestion

Involves enzymatic breakdown of nutrients

Different enzymes for proteins, carbohydrates, lipids

Occurs in mouth, stomach, small intestine

Enzymes specific to each nutrient type

Absorption Mechanisms

1. Gastrointestinal Absorption

Small intestine primary site of nutrient absorption

Villi and microvilli increase absorption surface

Active and passive transport mechanisms

Carrier proteins facilitate nutrient movement

2. Absorption Methods

Simple diffusion

Facilitated diffusion

Active transport

Endocytosis

Nutrient Functions

Carbohydrates

Primary energy source

Fuel for brain and muscle

Structural components

Metabolic regulation

Proteins

Structural components

Enzyme production

Hormone synthesis

Immune system support

Tissue repair

Energy production (secondary)

Lipids

Energy storage

Cell membrane formation

Hormone production

Insulation

Vitamin absorption

Inflammation regulation

Vitamins and Minerals

Metabolic cofactors

Antioxidant protection

Cellular signalling

Immune function

Bone health

Neurological processes

Absorption Locations

Mouth: Limited water, alcohol

Stomach: Minimal absorption

Small Intestine: Major nutrient absorption

Large Intestine: Water, electrolyte absorption

Factors Affecting Absorption

pH levels

Enzyme activity

Intestinal health

Nutrient interactions

Individual metabolism

Introduction

While proteins, carbohydrates, and fats are well-known nutrients, foods contain many other essential components that play crucial roles in human health and nutrition.

1. Phytonutrients

Antioxidant Compounds

Carotenoids: Found in yellow and orange fruits/vegetables

Flavonoids: Present in berries, tea, and chocolate

Polyphenols: Found in fruits, vegetables, and whole grains

Lycopene: Found in tomatoes

Anthocyanins: Found in blue and purple fruits/vegetables

2. Enzymes

Natural protein catalysts that facilitate biochemical reactions

Found in raw and fermented foods

Help with digestion and metabolic processes

3. Water

Often overlooked but critical nutrient

Essential for all bodily functions

Supports nutrient transportation, temperature regulation, and cellular processes

4. Probiotics and Prebiotics

Probiotics

Live beneficial bacteria

Found in fermented foods like yogurt, kefir, and kimchi

Support gut health and immune function

Prebiotics

Non-digestible fibres that feed beneficial gut bacteria

Found in foods like garlic, onions, and bananas

I. Digestion: Overview and Process

Digestion is a **complex** process of breaking down food into smaller molecular components that can be absorbed and utilized by the body. Digestion is a sophisticated, coordinated process involving mechanical and chemical transformations that enable nutrient absorption and metabolic functions.

Key Stages of Digestion

1. Mechanical Digestion

Physical breakdown of food through chewing, churning

Increases surface area for enzymatic action

2. Chemical Digestion

Enzymatic breakdown of food molecules

Occurs in mouth, stomach, small intestine

Involves specific enzymes for different nutrient types

II. Digestive System Anatomy

Primary Digestive Organs

Mouth

Oesophagus

Stomach

Small intestine

Large intestine

Accessory Digestive Organs

Liver

Pancreas

Gallbladder

III. Nutrient Absorption Mechanisms

Absorption Pathways

1. Passive Diffusion

Movement of molecules from high to low concentration

No energy required

2. Active Transport

Requires cellular energy (ATP)

Transport against concentration gradient

3. Facilitated Diffusion

Requires carrier proteins

No energy expenditure

IV. Nutrient Functions and Metabolism

Carbohydrates

Primary energy source

Stored as glycogen

Blood glucose regulation

Proteins

Structural components

Enzyme and hormone synthesis

Immune system support

Tissue repair

Lipids

Energy storage

Cell membrane formation

Hormone precursors

Thermal insulation

Vitamins and Minerals

Metabolic regulation

Enzyme cofactors

Antioxidant functions

Bone and cellular health

V. Key Absorption Sites

Small Intestine

Primary site of nutrient absorption

Specialized structures (villi and microvilli)

Maximizes absorption surface area

Large Intestine

Water reabsorption

Electrolyte balance

Bacterial fermentation

VI. Digestive Enzyme Functions

Carbohydrate Enzymes

Amylase (saliva, pancreas)

Breaks down complex carbohydrates

Produces simple sugars

Protein Enzymes

Pepsin (stomach)

Trypsin (pancreas)

Breaks peptide bonds

Produces amino acids

Lipid Enzymes

Lipase

Breaks triglycerides

Produces fatty acids and glycerol

B. Principles of Energy Balance

I. Definition of Energy Balance

Energy balance represents the relationship between energy intake (calories consumed) and energy expenditure (calories burned) in the human body.

II. Components of Energy Expenditure

1. Basal Metabolic Rate (BMR)

Energy required for basic physiological functions

Accounts for 60-75% of total daily energy expenditure

Influenced by age, sex, body composition, and genetics

2. Physical Activity Expenditure

Energy used during exercise and daily movements

Varies based on intensity, duration, and type of activity

Take 15-30% of total energy expenditure

3. Thermic Effect of Food (TEF)

Energy required to digest and metabolize nutrients

Approximately 10% of total daily energy expenditure

Differs across macronutrients (protein has highest TEF)

III. Energy Balance States

1. Positive Energy Balance

Calorie intake > Calorie expenditure

Results in weight gain

Excess energy stored as body fat

2. Negative Energy Balance

Calorie intake < Calorie expenditure

Results in weight loss

Body uses stored energy (fat, muscle) for metabolic functions

3. Neutral Energy Balance

Calorie intake = Calorie expenditure

Maintains current body weight

Optimal for weight maintenance

IV. Factors Influencing Energy Balance

Physiological Factors

Metabolism

Hormonal regulation

Muscle mass

Age and gender

External Factors

Diet composition

Physical activity levels

Sleep patterns

Stress

V. Energy Balance Calculation

Total Daily Energy Expenditure (TDEE) =
BMR + Physical Activity + Thermic Effect of Food

VI. Practical Implications

Weight management

Athletic performance

Nutritional strategies

Metabolic health

A. INTRODUCTION

Nutrition is a fascinating science that explores how food provides essential substances our bodies need to maintain health, grow, and function optimally. At its core, nutrition is about understanding the complex interactions between the foods we consume and our body's biological processes. **Essential Nutrients:** Nutrients are substances in food that the body needs for growth, repair, and energy.

Carbohydrates are our primary energy source, found abundantly in foods like grains, fruits, vegetables, and legumes. When we consume carbohydrates, the digestive process begins in the mouth with salivary enzymes and continues in the small intestine, where they are broken down into simple sugars that our body can easily absorb. While carbohydrates are essential, excessive consumption can lead to potential health issues such as obesity, insulin resistance, and metabolic disorders.

Proteins are the building blocks of life, critical for constructing and repairing body tissues. We obtain proteins from diverse sources including meats, eggs, dairy products, legumes, and nuts. The protein digestion journey starts in the stomach with pepsin and continues in the small intestine, where complex proteins are broken down into individual amino acids. These amino acids are fundamental for numerous bodily functions, including enzyme production, hormone regulation, and immune system support.

Fats, or lipids, often **misunderstood**, are actually vital for our health. They serve multiple functions beyond **energy storage**, including **hormone production**, **cell membrane formation**, and **protecting our organs**. Found in oils, butter, fatty fish, nuts, and various animal products, fats are metabolized through a complex process involving bile and specialized enzymes. However, it's important to understand that while fats are essential, an imbalanced intake can contribute to cardiovascular diseases and other health complications.

Beyond macronutrients, **micronutrients** play equally important roles. **Vitamins and minerals**, though required in smaller quantities, are critical for maintaining various physiological processes. Vitamins are categorized as fat-soluble (A, D, E, K) and water-soluble (B-complex, C), each supporting different bodily functions like metabolism, growth, and immune response. Minerals such as calcium, iron, zinc, and potassium are crucial for bone health, nerve transmission, muscle function, and numerous enzymatic activities.

Energy balance is another fundamental principle of nutrition. The concept is beautifully simple yet profoundly complex: the relationship between calories consumed and calories expended determines our body weight and metabolic health. Factors influencing this balance include basal metabolic rate, physical activity, diet-induced thermogenesis, and individual genetic variations.

Understanding nutritional needs requires recognizing that these needs vary across different life stages and individual circumstances. Children, elderly individuals, pregnant women, athletes, and those with chronic conditions all have unique nutritional requirements. A one-size-fits-all approach to nutrition is neither practical nor scientifically sound.

Practical nutritional guidance often centres on several key principles: consuming a diverse range of foods, practicing portion control, ensuring adequate hydration, and maintaining a balanced diet. It's not about extreme restrictions but about creating sustainable, healthy eating habits that support long-term well-being.

The field of nutrition is dynamic and continuously evolving. Ongoing research constantly unveils new insights into how different nutrients interact, their impact on our health, and the intricate ways our bodies process and utilize food. This makes nutrition an exciting and crucial area of study with direct implications for individual and public health.

B. ADDITIONAL NUTRIENTS AND COMPONENTS IN FOOD

In the realm of nutrition, understanding food goes far beyond the basic macronutrients of proteins, carbohydrates, and fats. A comprehensive exploration reveals a rich landscape of essential nutrients that play critical roles in human health and physiological functioning.

Phytonutrients represent a fascinating class of compounds found in plant-based foods. These **antioxidant compounds** including carotenoids, flavonoids, polyphenols, lycopene, and anthocyanin offer protective benefits and are abundant in fruits, vegetables, and whole grains.

Enzymes, natural protein catalysts found in raw and fermented foods, facilitate critical biochemical reactions and aid digestion and metabolic processes. **Water**, often overlooked, remains a fundamental nutrient **essential for all bodily functions**, supporting **nutrient transportation and cellular processes**.

The emerging understanding of gut health has highlighted the importance of probiotics and prebiotics. Probiotics are live beneficial bacteria found in fermented foods like yogurt and kimchi, while prebiotics are non-digestible fibre that nourish these beneficial bacteria, typically found in foods like garlic, onions, and bananas.

By recognizing the complexity and diversity of nutrients beyond macronutrients, we develop a more holistic approach to nutrition. A diverse diet ensures comprehensive nutrient intake, with each compound playing a unique and irreplaceable role in maintaining optimal health.

Fibre

Dietary fibre represents complex carbohydrates that cannot be digested by human enzymes. Found predominantly in plant-based foods like fruits, vegetables, whole grains, and legumes, fibre plays critical roles in digestive health. Soluble fibre helps regulate blood sugar and cholesterol levels, while insoluble fibre promotes regular bowel movements and prevents constipation. Different types include cellulose, hemicellulose, pectin, and resistant starch, each contributing uniquely to nutritional and physiological functions.

Emulsifiers

Emulsifiers are substances that help mix ingredients that typically don't combine easily, like oil and water. They stabilize food mixtures by reducing surface tension between different components. Commonly found in processed foods, natural emulsifiers include lecithin (from egg yolks and soybeans), mustard, and certain proteins. In food production, emulsifiers improve texture, extend shelf life, and enhance the overall sensory experience of products like mayonnaise, chocolate, and baked goods.

Pigments

Food pigments are natural or synthetic colouring compounds that determine the visual appearance of foods. Natural pigments include chlorophyll (green in vegetables), carotenoids (yellow-orange in carrots), anthocyanin (red-purple in berries), and betalains (red in beetroot). These compounds not only provide visual appeal but often serve as antioxidants with potential health benefits. Synthetic food colourings are also used extensively in processed foods to enhance aesthetic appeal and maintain consistent product appearance.

Flavour Compounds

Flavour represents a complex interaction of taste and aroma molecules that create sensory experiences in food. These compounds originate from various chemical structures including esters, aldehydes, ketones, and terpenes. Natural flavour sources include essential oils, herbs, spices, and fermentation processes. Umami, sweet, sour, salty, and bitter taste receptors interact with these molecules, creating the nuanced taste profiles we experience. Flavour development occurs through cooking processes like caramelization, Maillard reaction, and enzymatic breakdown, which transform raw ingredient characteristics.

Nutritional Significance

These additional food components: fibre, emulsifiers, pigments, and flavour compounds extend beyond basic nutritional roles. They contribute to the physical properties of food, sensory experience, and potential health benefits, demonstrating the complexity of nutritional science and food composition.

C. NUTRIENT DEFICIENCY DISEASES

Protein Deficiency Diseases manifest in two primary forms. Kwashiorkor is characterized by oedema, muscle wasting, and stunted growth, while marasmus represents a severe condition of calorie and protein deficiency that leads to extreme weight loss and muscle breakdown.

Vitamin Deficiencies can cause a range of serious health problems. Vitamin A deficiency results in night blindness and impaired immune function. Thiamine (Vitamin B1) leads to beriberi and neurological complications. Niacin (Vitamin B3) causes pellagra, which produces dermatitis, diarrhoea, and dementia. Vitamin B12 deficiency results in pernicious anaemia and neurological damage. Scurvy, caused by Vitamin C deficiency, leads to poor wound healing and weakened connective tissue. Vitamin D deficiency causes rickets in children and osteomalacia in adults.

Vitamin E deficiencies are rare but can cause neurological problems, while Vitamin K deficiency impairs blood clotting.

Mineral Deficiencies also have significant health impacts. Iron deficiency causes anaemia and reduces oxygen transport. Iodine deficiency leads to goitre and impaired thyroid function. Calcium deficiency results in osteoporosis and weak bones. Zinc deficiency causes growth retardation and impaired immune function. Magnesium deficiency can lead to muscle weakness and cardiac issues.

Primary risk factors for these deficiencies include poverty, limited diet diversity, malnutrition, and chronic health conditions.

D. DIGESTION, ABSORPTION AND FUNCTIONS OF NUTRIENTS

Digestion is a complex physiological process that transforms consumed food into nutrients essential for human body functioning. The journey begins in **the mouth**, where **mechanical breakdown through chewing** and **chemical breakdown via salivary enzymes** initiate nutrient processing. As food travels through the digestive tract, it undergoes progressive decomposition facilitated by specialized enzymes in different organs.

In the stomach, **gastric acids and pepsin** start protein degradation, creating **an acidic environment** that breaks down complex food structures. **The small intestine** becomes the primary site of nutrient absorption, featuring specialized structures called **villi and microvilli** that maximize surface area for efficient nutrient uptake.

Carbohydrates are primarily broken down into simple sugars like **glucose**, serving as the body's **primary energy source**. **Proteins** are decomposed into **amino acids**, crucial for **tissue construction, enzyme production, and immune system support**. **Lipids** are transformed into **fatty acids and glycerol**, essential for **energy storage, cell membrane formation, and hormonal processes**.

The **absorption** mechanisms involve **three primary pathways**: passive diffusion, active transport, and facilitated diffusion. These mechanisms enable nutrients to cross intestinal walls and enter the bloodstream, where they are transported to various body systems for metabolic functions.

Accessory digestive organs like the liver, pancreas, and gallbladder play critical roles in producing digestive enzymes, **neutralizing stomach acids**, and **supporting fat metabolism**. The **large intestine** primarily focuses on **water reabsorption** and **electrolyte balance**, **completing** the digestive process.

Vitamins and minerals, though required in smaller quantities, are fundamental in regulating metabolic processes, supporting enzymatic functions, and maintaining cellular health. Their absorption occurs **throughout the digestive tract**, with **different nutrients** having **specific absorption sites and mechanisms**.

The entire digestive system represents a sophisticated, interconnected network that transforms food into usable energy and structural components, supporting the body's continuous physiological functions.

E. ABSORPTION LOCATIONS AND INFLUENCING FACTORS IN NUTRIENT PROCESSING

Absorption locations represent a complex journey through the digestive system, with each segment playing a unique role in nutrient uptake. The mouth initiates minimal absorption, primarily for water and certain small molecules like alcohol. As food travels through the digestive tract, the stomach serves more as a processing chamber, with limited direct absorption.

The small intestine emerges as the primary nutrient absorption hub. Its extensive surface area, created by villi and microvilli, maximizes nutrient extraction efficiency. Different segments of the small intestine specialize in absorbing specific nutrients: the duodenum focuses on iron and calcium, the jejunum handles most carbohydrates and proteins, while the ileum specializes in vitamin B12 and bile salt absorption.

Factors affecting absorption are multifaceted and intricate. pH levels critically influence enzyme activity and nutrient solubility. Acidic or alkaline environments can dramatically alter nutrient availability and transportation mechanisms. Enzyme presence and functionality determine how effectively nutrients are broken down and prepared for absorption.

Individual metabolic variations significantly impact absorption rates. Genetic factors, gut microbiome composition, and overall digestive health modify nutrient uptake efficiency. Certain medical conditions like Crohn's disease, celiac disease, or chronic inflammation can substantially reduce absorption capabilities.

Nutrient interactions present another critical absorption factor. Some nutrients enhance each other's absorption, while others compete or inhibit uptake. For instance, calcium can impede iron absorption, and vitamin C enhances iron bioavailability. Dietary components like fibre, fats, and proteins can either facilitate or hinder nutrient absorption.

Physiological state, including age, hormonal balance, and overall health, influences absorption mechanisms. Stress, medications, and chronic conditions can alter digestive enzyme production and nutrient transport efficiency. The body's adaptive mechanisms continuously modulate absorption processes to maintain nutritional homeostasis.

F. ENERGY BALANCE

Energy balance is a critical metabolic concept describing the relationship between energy intake and expenditure. It represents how the body manages calories from food and drink against calories burned through physiological processes and physical activities.

The body's energy expenditure consists of three key components: **Basal Metabolic Rate (BMR)**, **physical activity expenditure**, and **the thermic effect of food**. BMR dominates energy use, representing calories required for fundamental life-sustaining functions. Physical activity

expenditure fluctuates based on movement levels, while the thermic effect of food accounts for energy needed to digest and metabolize nutrients.

Three primary energy balance states exist: **positive, negative, and neutral**. Positive energy balance occurs when calorie intake exceeds expenditure, resulting in weight gain. Negative energy balance happens when calorie expenditure surpasses intake, leading to weight loss. Neutral energy balance represents a stable state where energy intake precisely matches expenditure.

Numerous physiological and external factors influence energy balance, including metabolism, hormonal regulation, muscle mass, age, gender, diet composition, physical activity, sleep patterns and stress.

Tutorial Questions

1. Define nutrition and explain its importance in maintaining human health.
2. Describe the process of carbohydrate digestion, from initial breakdown in the mouth to final absorption in the bloodstream.
3. Compare and contrast fat-soluble and water-soluble vitamins, providing examples of each and their primary functions.
4. Explain the energy balance equation and discuss three factors that can influence caloric expenditure.
5. List and briefly describe the potential health problems associated with excessive intake of carbohydrates.
6. A 35-year-old athlete complains of frequent fatigue. Analyse potential nutritional factors that might contribute to her condition, considering macronutrient and micronutrient perspectives.
7. Explain how protein digestion occurs in the human body, detailing the roles of different enzymes and the final absorption mechanism.
8. Discuss the critical functions of lipids in the human body beyond energy storage, providing scientific explanations for each function.
9. Compare the nutritional needs of a child and an elderly person, highlighting the differences and the importance of tailored nutritional approaches.
10. Analyse the statement: "All fats are harmful to health." Use scientific evidence to construct a nuanced response that explains the complex role of lipids in nutrition.
11. Define energy balance and explain its three primary states.
12. Describe the components of total daily energy expenditure and their relative contributions.
13. Analyse how hormonal systems regulate energy balance and metabolic processes.

14. Calculate the total daily energy expenditure (TDEE) for an individual, considering age, weight, height, and activity level.
15. Discuss potential health consequences of prolonged positive or negative energy balance.
16. Compare energy balance dynamics in sedentary versus active populations.
17. Evaluate the impact of macronutrient composition on metabolic rate and energy balance.
18. Design a personalized energy balance strategy for weight management.